Artificial Intelligence Applications for Defect Detection in Industrial Processes: a Bibliometric Analysis

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Abstract. Artificialintelligencealgorithms,especiallymachinelearningand deeplearningtechniques,areincreasinglypresentinourdailylives,often presentingperformanceevolutionsandnewapplications.Withtheadventof newtechnologiesandtheso-calledindustry4.0,applicationsinindustrialand manufacturingprocessesareincreasingly**present. This work aims to stpiulate a** theoreticalscope and trends regarding the works that emphasize AI applications inindustrialandproductionprocesses.Abibliometricanalysiswas carried out (usingVOSViewersoftware)onscientificpublicationslinkedtoterms such as machineanddeeplearning,associatedwithapplicationsinfailureanderror detectionof products and procedures in industrial and manufacturing processes. Thestudyreviewedpublicationsfrom2015to2022inScopusandWebof Science platforms.

Keywords: Deep Learning, Machine Learning, Bibliometric analysis.

1 Introduction

Artificialintelligence(AI)canbedefinedastheefforttoautomateintellectualtasks performedbyhumans.AIisabroadfieldthatencompassesmachinelearning(ML) techniquesanddeeplearning(DL)techniques,butitisnotlimitedtoML and DL. It alsohasapproachesthatdonotdependonlearning, such as clustering techniques [1] and anomaly detection [14].

Therearecountlesspossibleapplicationsforthevariousexistingaltechniques, manyofthempresentinthe daily lives of the vast majority of the world's population, suchas product, film, and series recommendations systems, spam recognition systems ine-mails,facialrecognitionsystemsonsmartphones,andothers.Oneofthe applicationareasthathasbeengainingmoreandmorerelevanceinseveral internationalscenarios(Germany,UnitedStates,China,andJapan)isfocusedon

intelligentmanufacturingsystems[2].AccordingtoXuandHua(2017),these systemshaveastheirmaincharacteristics*highcorrelationanddeepintegration, dynamicreconfiguration,andahugevolumeofdata.* In terms of high correlation, all systemsarehighlycorrelated,exchanginginformationindifferentlayersofthe process, such as manufacturin and storage/ systems exchanging information.Indynamicreconfiguration,allsystemsareconnectedtocreatethe best strategiesfordifferentproducts/servicesautomatically.Finally,intermsofthelarge volumeofdata,eachexisting"intelligent"systemgeneratesahugeamountofdata fromthemostdiversepossiblesourcestobeusedformonitoring,transmission, processing, and storage.

Techniqu encompass by AI can be implement at practically all manufacturingsystemlevels,suchasincrucialpartsthatdeterminethequalityand successofaproduct,suchasfaultdetection[3,4]andaiding manual assembly steps [5,15].Thammetal.(2021)haveproposedanaugmented intelligence-based worker assistance system in manual assembly.

Thisworkaimstoperformabibliometricanalysisofpublicationsrelated to terms associatedwithArtificialIntelligence,suchas Machine Learning and Deep Learning. Weassociatedthemwithothertermsthatshowapplicationsinfailureanderror detection,productsandproceduresandindustrialandmanufacturingprocesses.Our purposewastostipulateatheoreticalscopeanddiscovertrendsregardingresearch beingcarriedout,aswellasidentifymainauthors,publications,countriesandall relevantdataforthischaracterization,emphasizingAIapplicationsinindustrial processes.Itconsiderspublicationsfrom2015to2022intheScopusandWebof Science databases. The data obtained were analyzed in terms of various characteristics withtheaidoftheVOSViewersoftwareforbibliometricanalysis.Possible applications,casestudies,trends,andresearchgapsonthetopicsinquestionwere identified.

2 Methodology

Bibliometricsanalysiscanbedefinedastheuseofmathematicalandstatistical methodstoshowquantitativeandqualitativechangessurroundingacertainareaof researchtostipulateimportantaspectsinthisarea,suchastrends,mainworks, authors, and others [13].

Theadvancedbibliometric analysis was based on the steps defined and specified in this section.

2.1 Databases

Initially,thedatabaseswereselectedtosearchforscientificarticles. In this work, the databasesusedwereScopusandWebofScience.Theyareplatformswithlarge

databasesofreviewedscientificpublicationscontainingworkssuchasbooks, scientificjournals,andconferenceproceedingsfromthemost diverse scientific areas (technology,medicine,social sciences, arts, and humanities), in addition to presenting several search tools and filters.

2.2 Research Protocol

Intheformthattheresultsobtainedpresentthemostrelevantand necessary data for theobjectiveinquestion,aresearchprotocolwasdefinedtorelatekeywordswith Booleanoperators(ORandAND)togeneratesearchsentences.Thesesearch strategieswerelaterappliedontheplatformsintheirrespectivesearchsystemsin keywords, abstracts, and titles of publications. The search phrase used was:

((recognitionAND(defectORfail))AND((assemb* OR manufact*) AND (lines OR production)) AND ("deep learning" OR "machine learning" AND algorithm))*

Itrelatesaspectsoffaultdetectioninproduction lines from AI techniques, and the *characterisinterpretedasawildcard,accepting any combination of letters after the last term. It was done to cover more related works.

Thirty-eightpublicationswereobtainedintheScopusdatabaseand58in the Web ofSciencedatabase.Still,asbothbasessharejournalsincommon,somesamples werepresentinbothgroupsofresults,soitwasnecessarytoeliminateduplicate worksinthesample,obtainingthen16worksintheScopusdatabaseand58inthe Webof**Science database, totaling 74 publications. It was also defiend that only works** writteninEnglishwouldbeusedinthe sample, which made it necessary to eliminate onemorepublicationpresentinthesamplefromtheScopusbase,becauseitwas writteninChinese,thustotaling73publications.Table1summarizestheresearch protocol used.

Table 1. Publication selection protocol.

Methodological Procedures	
Database	Scopus and Web of Science
Publishing period Boolean operators Publication types	2015 to 2022
	“AND”, “OR” and “*”
	Articles, conference papers
Keywords Combination	
Textual content Terms	TITLE-ABS-KEY-AUTH
	"recognition", “defect”, “fail”, “assemb*”, ”algorithm”, "production”, "deep learning", “machine learning”Final combination TITLE-ABS-KEY-AUTH ((recognition AND (defect OR fail*)) AND((assemb*ORmanufact*)AND(linesORproduction))

AND ("deep learning" OR "machine learning" AND algorithm))

Portfolio of References Selected for Quantitative Analysis

Publications found and used in the quantitative analysis:	96 results
Elimination of duplicate publication (present in both bases):	22 results
Elimination of publications in languages other than English and Portuguese:	1 result
Final portfolio of publications	**73 results**

2.3 Bibliometric Data

Withthelistofselectedpublications,itispossibletoexportvariouspublication informationfromthedatabases,suchasthenumberofcitations,yearofpublication, thecongressinwhichitwaspublished,andvariousotherinformation.In possession ofthesesamples,itispossibletoevaluatesomedatathatdescribe the characteristics oftheresearchareasunderanalysis.Theopen-sourcesoftware VOSviewer was used tobuildandvisualizebibliometricnetworks.Itpermitsus to analyze the data present inthesample,whichcanbeformedbycategoriessuchasauthors,journals, publications,andothers,whichcanbeinterrelatedbasedoncitations,bibliographic coupling, co-citation, and co-authorship. These data are further explored in section 3.

3 Result and Discussion

3.1 Bibliometric Analysis

Thissection presents and discusses the results obtained from the procedures described above.Fromthe73articlesobtained with the defined research protocol applied to the ScopusandWebofSciencedatabases,itwaspossibletoobtainrelevantdatato quantifythematurityandcharacteristicsoftheresearcharea.Weidentifiedstudied themes,suchasmainauthors,countrieswithmorepublications,moreimpactful works, and others.

Thegraphbelow(Fig.1)showsthenumberofworkspublishedinthedefined period.Thegrowthtrendinthesepublicationsovertheyearsisclear.Itcanbe characterizedasevidencethatthe scientific community increasingly works on studies relatedtothetopicsinquestion.Theyear2022hadthehighestnumber,with32

publications.Anincreaseof68.42%over2021,whilethenumberofpublicationsin 2021 corresponds to an increase of 58.33% in 2020.

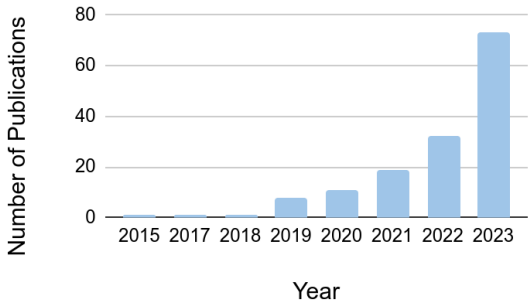

Fig. 1. Number of publications between 2015 and September 2022.

Table2presentsthe nine most cited works among the 73 works present in the sample, togetherwiththeirJournalCitationReports(JCR)index,whichwascreatedto evaluatetherelevanceofajournal,takingintoaccountdatasuchasthenumberof citations and papers in a journal [6].

Table 2. Most cited works, among the 74 present in the sample.

Authors	Title	Citations	Journal	JCR
Bertolini, M. et al. (2021)	Machine Learning for applications: A literature review	**58**	Expert Systems with Applications	**8.665**
Tan, SC. et al. (2015)	Evolutionary Fuzzy Neural Networks for Classification of Defects	**46**	IEEE Transactions on Neural Networks and Learning Systems	**14.255**
Wang, JL. et al. (2020)	Deformable Networks for Efficient Mixed-Type Wafer Recognition	**34**	IEEE Transactions on Semiconductor Manufacturing	**2.796**
Yang, YT. et al (2020)	A High-Performance Deep Learning Algorithm Automated Optical Laser Welding	**33**	Applied Sciences	**2.838**

Gao, YP. et al (2020)	A Multilevel Fusion-Based Deep Method for Vision-Recognition	**26**	IEEE Transactions on Neural Networks and Learning Systems	**12.255**
O'Leary et al. (2020)	Deep Learning for of the Chemical Particle Defects on semiconductor Wafers	**22**	IEEE Transactions on Semiconductor Manufacturing	**2.796**
Gao, YP et al. (2022)	A Review on Recent Vision-based Defect towards Industrial Intelligence	**20**	Journal of Manufacturing Systems	**8.498**
Xu, L. et al. (2020)	A Weakly Supervised Defect Detection Convolutional Neural	**19**	IEEE Acess	**3.476**
Bahaghighat, M; Akbari, L; Xin, Q. (2019)	A Machine Learning-Approach for Cards Within Drug Packages	**15**	IEEE Acess	**3.476**

Althoughthemostcitedpublicationinthesamplewasnottheoldest,letalonethe workpublishedinthejournalwiththehighestJCRindex,thisnumber of citations is directly linked to the fact that this work is a literature review.

Intendingtoidentifythecountriesthatpublishthemostworksonthetopicsin question,the graph in Fig. 2 presents the number of publications per country, in which publicationsin25countrieswereidentified.Chinaholdsthelargestnumberof publications,about35.71%ofpublicationsinthestipulatedperiod,followedby Taiwan(10.71%),Germany(9.52%),andtheUSA(5.95%),inthegraph,"others" represent countries that contain only one publication.

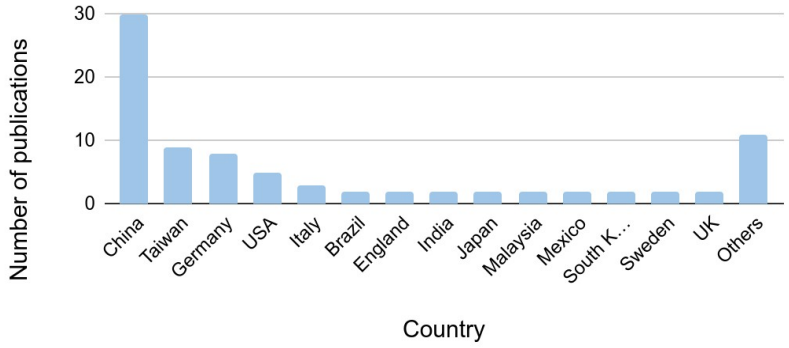

Fig. 2. Number of publications per country.

Thekeywordanalysispresentsuswithpossibletrendsforfuturestudiesby connectingwithotherunusedkeywordsinoursearchprotocol.Itwaspossibleto correlatesomedatapresentinthepublicationsinthesample,allowingusto identify themainreferences,authors,andtrends.Thekeywordco-occurrencemapinthe sampleof73paperscanbeseeninFig.3,inwhichkeywordsthatoccurinatleast two papers were selected, totaling 79. Table 3 presents the most frequent among them, with the keyword *deep learning* at the top of the list, representing about 34.17% of the total.It also highlights *convolutional neural networks* (and their similarities like *CNN, convolutio neural network,* etc.) representing 27.84% and the keyword *classification* (20.25%).

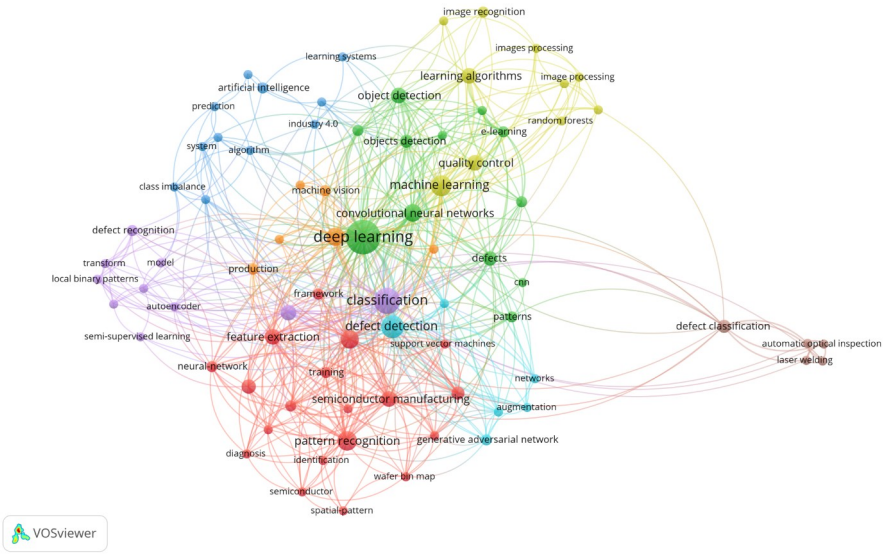

Fig. 3. Obtained keyword occurrence map.

Table 3. Most frequent keywords in the sample.

Keyword	Occurrence
Deep learning	27
Convolutional neural networks Classification	22
Defect detection	16
Machine Learning	12
Pattern recognition	10
Inspection	9
Recognition	8
Feature extraction	8
	6

Thekeywordsco-occurrencemapiscomposedof6mainclusters,withthemain keywords*deeplearning*(greencluster),*cla*ssification(purplecluster),*defect detection*(bluecluster),*patternrecognition*(redcluster),*machinelearning*(yellow cluster),and*defectdetection*(browncluster).Byanalyzingtheseclusters,wecan identifythemostusedtermsandhowtheyrelatetoeachothertoidentifythemain inspirationsandtrendsfortheareasthatarebeinganalyzed,suchasthekeywords *augmentation*,*generative adversarial network* (GAN), and *extraction features* present intheblue cluster,linkedtotermssuchas*clustering algorithms*, *data augmentation,* and*training*thatcandemonstratethe use of deep learning architectures such as GAN togeneratedataforsupervisedAIalgorithms,aswellastheuseofclustering techniquesforpatternrecognition.Anothercasethatcanbeobservedis the presence ofthekeywords*surface defect recognition* and *semi-supervised learning* linked to the terms*classification*,*defectdetection,*and*qualitycontrol*,whichcanbeseenas possible application trends.

OthermapsthatcanbecreatedfromVOSVIewerarethebibliographiccoupling andco-citation maps. Bibliographic coupling establishes associations between articles basedonbibliographicreferencespresentinbotharticles,whichmaytake publicationsorauthorsintoaccount.On the other hand, co-citation associates two (or more)documentsorauthorsbasedonthepresenceofthesesamedocuments/authors in the same list of references of a publication [8].

Theminimumnumberofdocumentspresentsimultaneously in two or more works inoursampletobeconsideredacouplingwasdefinedaccordingtothePareto principle.Sothat20%ofthe 73 publications obtained were represented. Fig. 4 shows thebibliographiccouplingmapobtained,inwhich15(approximately20%ofour sample)linkeddocuments were obtained. A point to be noted is that all the most cited publicationsinoursampleareamongthese13documents,evenjustthreebeing literaturereviewsonthestudiedthemes.Thisanalysisallowsustoassimilatethe proximity of publication from their references on the theme, theoretica

methodologicalnature,oranothersimilarsharedaspect[8,9].Itisalsopossibleto notethattheworksbyBertolini,Metal.(2021)andGaoYP et al. (2022) are linked tothemainclustersonthemap,coveringalmosttheentiremap,whichcanbegood evidence that these works contain good references for the topics in question.

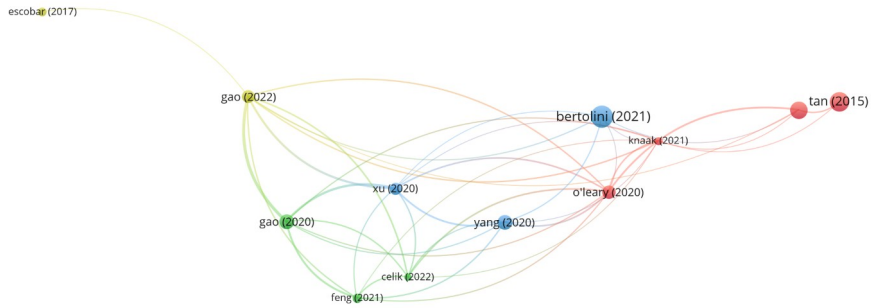

Fig. 4. Obtained bibliographic coupling map.

Forcreatingtheco-citationmapofauthors,itwasdefinedthattobeconsidereda co-citation, two or more authors must be cited simultaneously in about 20% (using the Paretoprinciple)ofthepublicationsinoursample.Wethenobtainedsevenco-cited authors.Themapobtainedcanbeseeninfig.5,where the main authors cited in our sampleandtheirrelationshipsarepresent,whileTable4showsthemostco-cited authors.

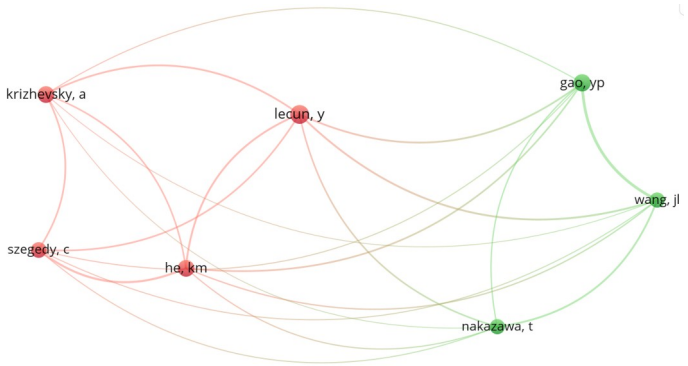

Fig. 5. Author co-citation map.

Amongthe most cited authors, it is possible to observe famous names in the academic areaonthesubjects,especiallyinthemachineanddeeplearning.Thatisthe case of Lecun,Y.,whoisresponsibleforpioneeringworksindeeplearning,suchasthe

creationofCNN(ConvolutionalNeuralNetwork)namedLeNet[10].He is the chief AI scientist on Facebook and Professor at New York University nowadays.

Table 4. Most co-cited authors are present in the co-citation analysis.

Author	
Lecun, Y.	18
Gao, Yp.	15
He, KM.	14
Krizhevsky, A. Szegedy, C.	14
Nakazawa, T. Redmon, J.	13
	12
	11

Atthesametime,Redmon,J.(researcherattheUniversityofWashington) is known forcreatinganobjectdetectionarchitectureoptimizedconcerninginferencetime, popularly known as YOLO [11]. This architecture continues to be studied and evolved yearafteryearnextto state the art for object detection tasks. Based on these data, it is possibletostatethattheseauthorsarepioneersinthe algorithms used by most of the workspresentinthesample,addressingthemostdiverseaspects,suchasnew architectures for different tasks, optimization methods, and the creation of datasets.

4 Conclusions

Thestudyidentifiedthatthenumberofpublicationsabouttheapplicationsof techniquesandartificialintelligencealgorithmstodetectfailuresinmanufacturing andindustrialprocessesissmall,evenwiththeresearchprotocolbeingstrictly specificandappliedtothetwolargestandmost frequently used platforms. However, itispossibletonoticeagrowingtrendintheobservedperiod,showingsignsof increasing research opportunities and gaps that are of interest.

Bibliometricanalysisalsorevealsrelevantinformation,suchasmajorcountries investinginresearchofthissubject,i.e.,ChinaandtheUnitedStates.It is important toemphasizethatthelownumberofpublicationsinSouthAmericamayreflectthe industrycrisisthatoccurredintheregionduringthetimeperiod.inwhichallworks publishedinthisperiodareofBrazilianorigin,makingthenationalresearchand development of works related to the theme of this paper even more relevant.

ItisalsoimportanttoconsiderdataanalyzedthroughVOSViewer,whichallows significantanalysisforfutureworkssincethemostrelevantresearcherswerefound throughthenumberofcitationsandco-citationsoftheirpublisheddocuments. Moreover,weidentifiedthemainkeywordsandtheirrelationships,which present us

withthemainareasandsubareasonthesubject.Thiscanbeobservedwhenwe analyzethemostcitedkeywordsinthesamplealongwiththemainauthors.That analysis highlighted Deep LearningandConvolutional standpointsforfuturestudies.Italsoallowusto identify that the keywords related to recognitionanddetectionarethe most present in a general context, which are directly linked to the main authors presented in this work.

Inspiredbytheresultsofthisresearch,theauthorsareinvestigatingthe developmentofasystemcapableofdetectingfailuresinmanualassembly processes inindustrialenvironment.Inaddition,aqualitativeanalysiscanbeperformed on the publicationsinvestigatedinthisresearchto further explore aspects of the main works onwhichthesethemescanbefurtherinvestigated,generatingevenmoreresearch insightsandthedevelopmentofananalyticalframeworkforevaluatingnew technologies in this field.

Finally,itispossibletostatethatthisworkvalidatestherelevanceandgrowthof futurestudiesonthesetopics,facilitatingresearchdirections,providingquality references,andfosteringrelevantinsightsinacademia,inordertopresentthemain techniques, algorithms and paradigms applied to tasks in question.

References

1. Chollet, F.: Deep Learning with Python. [S.l.]: Manning Shelter Island (2018).
2. Xu, X.. Hua, Q.: Industrial big data analysis in smart factory: Current status and research strategies, in IEEE Access, vol. 5, pp. 17 543–17 551, (2017).
3. Sassi,P.,Tripicchio,P.,andAvizzano,A.C.:Asmartmonitoringsystemforautomatic weldingdefectdetection,IEEETransactionsonIndustrialElectronics, vol. 66, no. 12, pp. 9641–9650, (2019).
4. Du,W.,Shen,H.,Fu,J.,Zhang,G.,He,Q.:Approachesforimprovementofthex-ray image defect detection of automobile casting aluminum parts based on deep learning, NDT E International, vol. 107, p.p 102144, (2019).
5. Lai,Z.-H.,Tao,W.,Leu,M.C.,Yin,Z.: Smart augmented reality instructional system for mechanicalassemblytowardsworker-centeredintelligentmanufacturing,”Journalof Manufacturing Systems, vol. 55, pp. 69–81, (2020).
6. JOURNALCITATIONREPORTS.Cartãode referência rápida: suportado pelo Isi Web of Knowledge: Thomson Reuters, (2009) Disponível em: <https://bibliotecaquimicaufmg2010.files.wordpress.com/2013/03/jcr_qrc_pt_20101026.pdf>.last accessed 2023/01/08.
7. Escobar,C.A.,Morales-Menendez,R.:MachineLearningandPatternRecognition TechniquesforInformationExtractiontoImproveProductionControlandDesign Decisions.In:Perner,P.(eds)AdvancesinDataMining.ApplicationsandTheoretical Aspects.ICDM2017.LectureNotesinComputerScience(),vol10357.Springer,Cham (2017).

8.Grácio, M. C. C.: Acoplame bibliográfi e análise de cocitação: revisão teórico-conceitual.EncontrosBibli:RevistaEletrônicadeBiblioteconomiaeCiênciada Informação, v. 21, n. 47, pp. 82-99, (2016).

9.Lucas,E.O.: Garcia-Zorita, J. C.: Sanz-Casado, E.: Evolução histórica de investigação em **informetria: ponto de vista espanhol. Licn em Revista, v. 9, n. 1, pp. 255-270, (2013).**

10.Lecun, Y., Bottou, L., Bengio, Y., Haffner, P.: Gradient-based learning applied to document recognition, in Proceedings of the IEEE, vol. 86, no. 11, pp. 2278-2324, (1998).

11.Redmon,J.,Divvala,S.,Girshick,R.,Farhadi,A.;YouOnlyLookOnce:Unified, Real-TimeObjectDetection,inProceedingsoftheIEEEConferenceonComputerVision and Pattern Recognition (CVPR), pp. 779-788 (2016).

12.Lihao,W.andYanni,D.:AFaultDiagnosisMethodofTreadProductionLineBasedon ConvolutionalNeuralNetwork,2018IEEE9thInternationalConferenceonSoftware Engineering and Service Science (ICSESS), pp. 987-990, Beijing, China (2018).

13.DeBakker,F.G.A.,Groenewegen,P.,&DenHond,F.:Abibliometricanalysisof30 yearsofresearchandtheoryoncorporatesocialresponsibilityandcorporatesocial performance. Business and Society, vol 44(3), pp 283-317 (2005).

14.Breunig,M.M.,Kriegel,H-P.,RaymondT.Ng,Sander,J.2000.LOF:identifying density-based local outliers. SIGMOD Rec. 29, 2 ,pp. 93–104 (2000).

15.Thamm,S.,Huebser,L.,Adam,T.,Hellebrandt,T.,Heine,I.,Barbalho,S.,KnorrV.S., Becker,M.,Bagnato,S.V.,Schmitt,H.R.:Conceptforanaugmented intelligence-based qualityassuranceofassemblytasksinglobalvaluenetworks,ProcediaCIRP,vol97,pp 423-428 (2021).

16.Bertolini,M.,Mezzogori,D.,Neroni,M.,Zammori,F.:MachineLearningforindustrial applications:Acomprehensiveliteraturereview,ExpertSystemswithApplications,vol 175, 114820 (2021).

17.Gao,Y.,Li,X.,Wang,X.V.,Wang,L.,Gao,L.:AReviewonRecentAdvancesin Vision-based Defect Recognition towards Industrial Intelligence, Journal of Manufacturing Systems, vol 62, pp 753-766, (2022).

18.vanEck,N.J.;Waltman,L.(2010)VOSViewer:VisualizingScientificLandscapes [Software]. Available from https://www.vosviewer.com